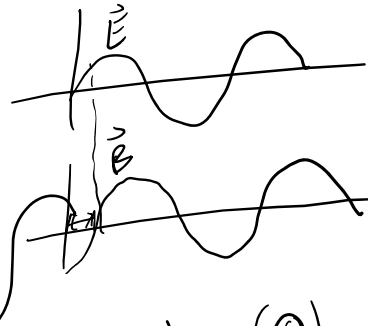


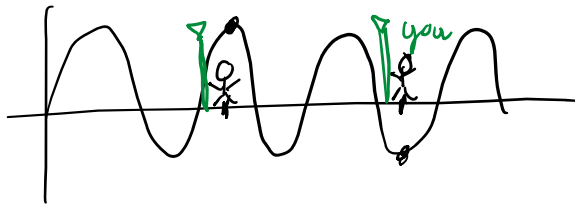
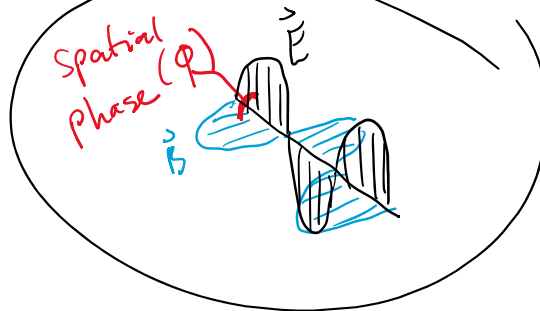
Lecture 2

$$\frac{\partial^2 \vec{E}}{\partial t^2} - \frac{1}{\mu_0 \epsilon_0} \nabla^2 \vec{E} = 0$$

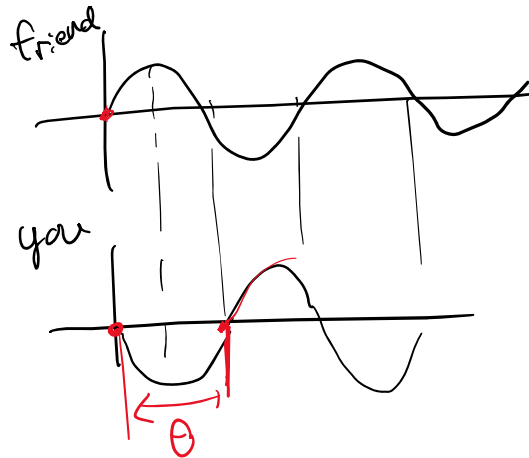
$$\frac{\partial^2 \vec{B}}{\partial t^2} - \frac{1}{\mu_0 \epsilon_0} \nabla^2 \vec{B} = 0$$

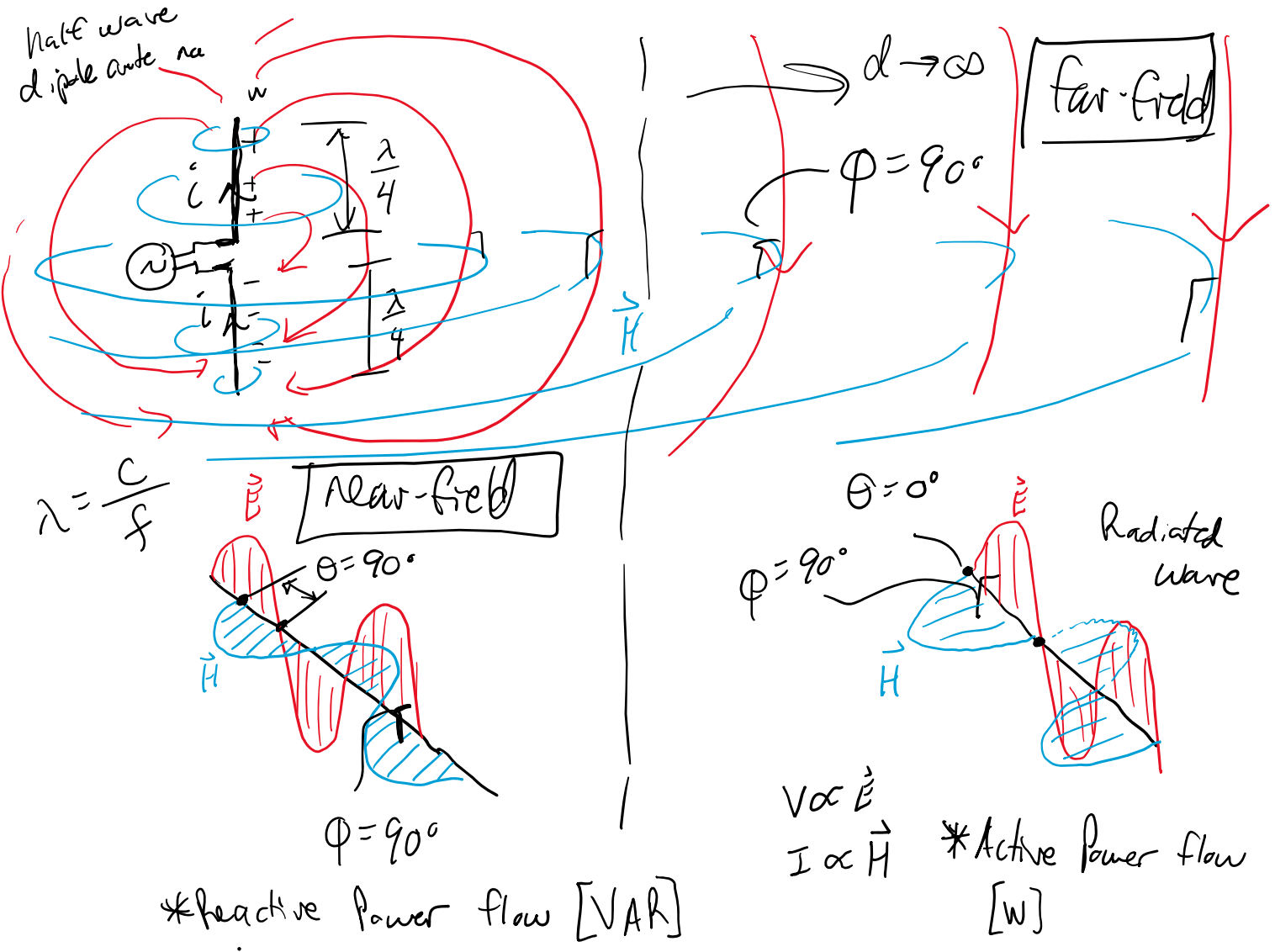


temporal phase (θ)

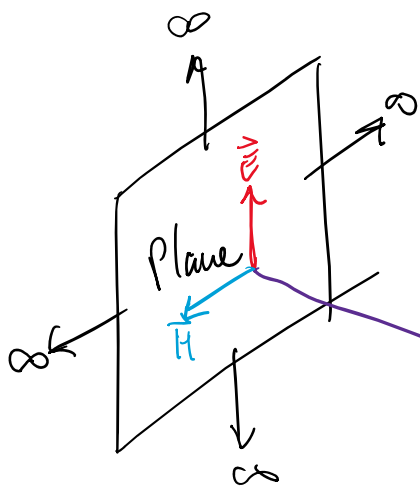


O-Scope would measure





Plane waves : Math trick to simplifying the equations for $\vec{E}$ and $\vec{H}$ in the far-field



* for this to work distance (d) from antenna (source) $d \gg x$, where x is the location of the antenna.

stays a plane

$|\vec{k}|$ propagation constant

time distance

$t = t_1$